The documentation and process conversion measures necessary to comply with this revision shall be completed by 4 June 2021.

INCH-POUND

MIL-PRF-19500/542K
W/AMENDMENT 1
4 March 2021
SUPERSEDING
MIL-PRF-19500/542K
23 October 2017

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD EFFECT, N-CHANNEL,
SILICON, ENCAPSULATED (THROUGH HOLE PACKAGE),
TYPES 2N6756, 2N6758, 2N6760, 2N6762, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel, enhancement-mode, MOSFET, power transistors. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device. Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device.

1.2 Package outlines. The device package outlines are as follows: TO-204AA; formerly TO-3 in accordance with figure 1 for all encapsulated device types. See figures 2 and 3 for unencapsulated devices.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T (1) T_C = +25°C	P_T T_C = +25°C (free air)	V_{DS}	V_{DG}	V_{GS}	I_{D1} (2) (3) T_C = +25°C	I_{D2} (2) T_C = +100°C
	W	W	V dc	V dc	V dc	A dc	A dc
2N6756	75	4	100	100	± 20	14.0	9.0
2N6758	75	4	200	200	± 20	9.0	6.0
2N6760	75	4	400	400	± 20	5.5	3.5
2N6762	75	4	500	500	± 20	4.5	3.0

Type	I_S	I_{DM} (4)	T_J and T_{STG}	V_{ISO} 100,000 feet altitude	Max $r_{DS(on)}$ (1) V_{GS} = 10 V dc, I_D = I_{D2}		$R_{\theta JC}$ max
					T_J = +25°C	T_J = +150°C	
	A dc	A (pk)	°C		ohms	ohms	°C/W
2N6756	14.0	56	-55 to +150		0.18	0.36	1.67
2N6758	9.0	36	-55 to +150		0.4	0.84	1.67
2N6760	5.5	22	-55 to +150	400	1.0	2.5	1.67
2N6762	4.5	18	-55 to +150	500	1.5	3.75	1.67

See notes on next page.

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1.3 Maximum ratings - Continued.

(1) Derate linearly 0.6 W/°C for T_C > +25°C;
(2) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(3) See figure 4, maximum drain current graph.
(4) I_{DM} = 4 x I_{D1} as calculated in note 2.

1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = 1 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		Max I_{DSS1} V_{GS} = 0 V	Max $r_{DS(on)1}$ (1) V_{GS} = 10 V dc I_D = I_{D2}
				V_{DS} = 80 percent of rated V_{DS}	T_J = +25°C
	V dc	V dc		µA dc	ohms
		Min	Max		
2N6756	100	2.0	4.0	25	0.18
2N6758	200	2.0	4.0	25	0.4
2N6760	400	2.0	4.0	25	1.0
2N6762	500	2.0	4.0	25	1.5

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.6 for PIN construction example and 6.7 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

1.5.2 JAN certification mark and quality level for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "6756", "6758", “6760”, and "6762".

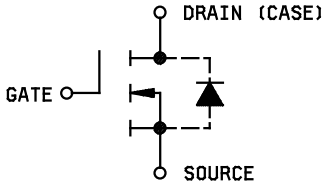
1.5.3.3 Suffix letters. No suffix letters are used on devices that are packaged in the TO-204AA (formerly TO-3) package of figure 1.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

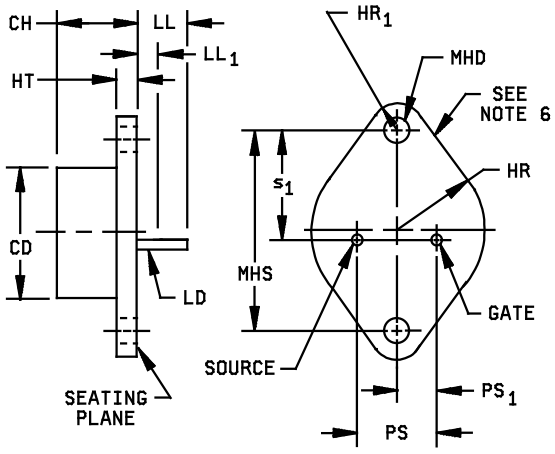
1.5.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A" and "C" (see figures 2 and 3 and 6.6).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.23	
CH	.250	.360	6.35	9.14	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	
HT	.060	.135	1.52	3.43	
LD	.038	.043	0.97	1.09	
LL	.312	.500	7.92	12.70	
LL_1		.050		1.27	
MHD	.151	.165	3.84	4.19	
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	3, 5
PS_1	.205	.225	5.21	5.72	3, 5
s_1	.655	.675	16.64	17.15	



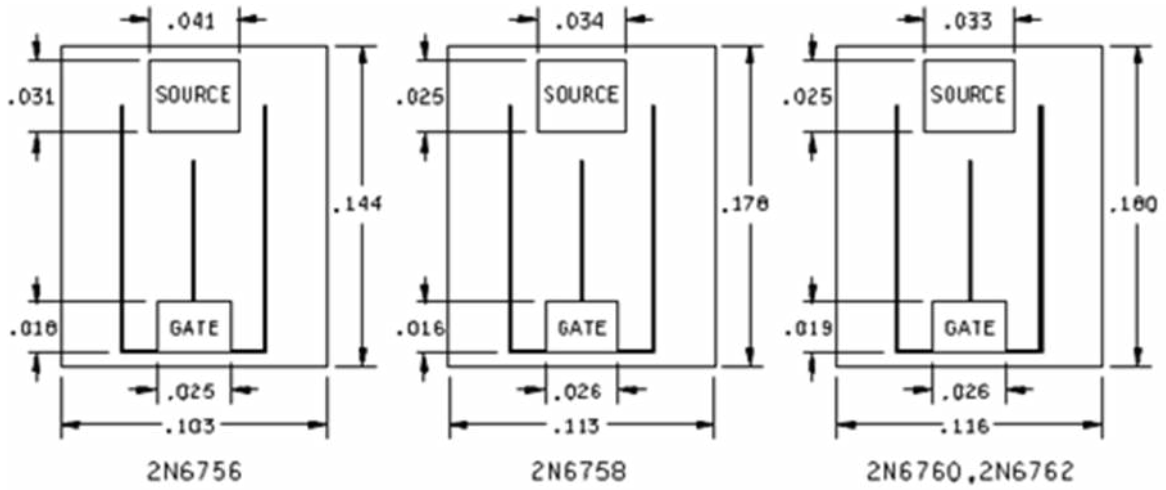
CIRCUIT SCHEMATIC



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) and .055 inch (1.40 mm) below seating plane. When gauge is not used measurement will be made at the seating plane.
4. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
5. Mounting holes shall be deburred on the seating plane side.
6. Drain is electrically connected to the case.
7. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions of transistor (TO-204AA).

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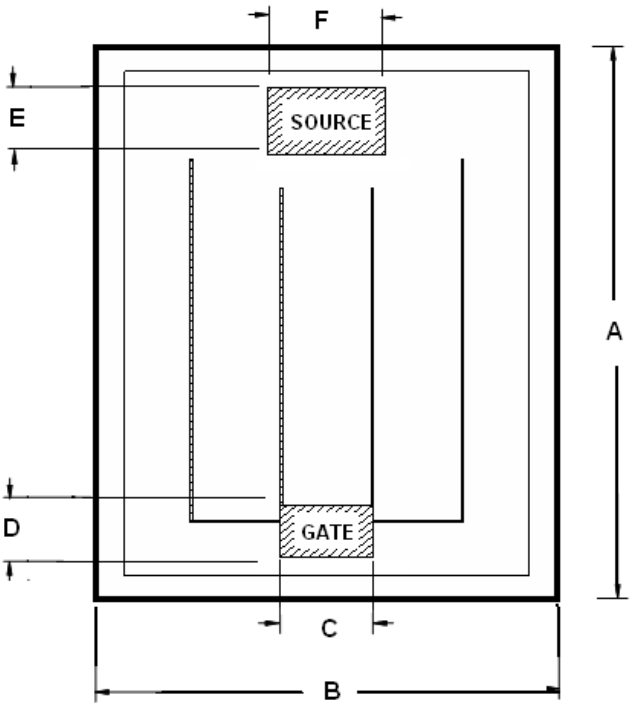


Inch	mm	Inch	mm	Inch	mm
.016	0.41	.031	0.79	.113	2.87
.018	0.46	.033	0.84	.116	2.95
.019	0.48	.034	0.86	.144	3.66
.025	0.64	.041	1.04	.180	4.57
.026	0.66	.103	2.62	.178	4.52

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is .005 inch (0.13 mm).
4. The physical characteristics of the die thickness are .0187 inch (0.475 mm). The back metals are chromium, nickel and silver. The top metal is aluminum and the back contact is the drain.
5. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

* FIGURE 2. JANHCA and JANKCA (A-version).

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Ltr	Dimensions - 2N6756				Dimensions - 2N6758			
	Inches		Millimeters		Inches		Millimeters	
	Min	Max	Min	Max	Min	Max	Min	Max
A	.181	.185	4.60	4.70	.179	.183	4.55	4.65
B	.116	.120	2.95	3.05	.114	.118	2.89	2.99
C	.032	.034	.81	.86	.028	.030	.71	.76
D	.017	.019	.43	.48	.018	.020	.46	.51
E	.024	.026	.61	.66	.024	.026	.61	.66
F	.035	.037	.89	.94	.033	.036	.84	.91

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are: The back metals are chromium, nickel, and silver and the back contact is the drain. The top metal is aluminum.
5. Die thickness is .015 inch (0.38 mm) ±.001 inch (0.025 mm).

FIGURE 3. JANHCC and JANKCC (C-version) die dimensions for 2N6756, 2N6758.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.
* MIL-STD-883 - Test Method Standard for Microcircuits.

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

nC ..... nano coulomb.
I_{AS} ..... Rated avalanche current, non-repetitive.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-204AA; formerly TO-3), figures 2, and 3 for JANHC and JANKC die dimensions.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction is not permitted to meet the requirements of this specification.

* 3.4.3 Silicone die coating. The use of a silicone die coat requires a successful completion of MIL-STD-883, method 5011 on each silicone lot for its intended applications, and as part of the full MIL-PRF-19500 qualification process.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

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* 3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices. shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ kΩ, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be table I as specified herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be as specified in MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500) (1) (2)	Measurement	
	JANS level	JANTX and JANTXV levels
(3)	Gate stress test (see 4.3.2)	Gate stress test (see 4.3.2)
(3)	Method 3470 of MIL-STD-750, (see 4.3.3) optional	Method 3470 of MIL-STD-750, (see 4.3.3) optional
(3) 3c	Method 3161 of MIL-STD-750, (see 4.3.4)	Method 3161 of MIL-STD-750, (see 4.3.4)
* 5	Method 2052 of MIL-STD-750, PIND (see MIL-PRF-19500 and 4.3.5)	Not applicable
9	I_{GSSF1}, I_{GSSR1}, I_{DSS1}	Not applicable
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table I herein: ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table I herein.
12	Method 1042 of MIL-STD-750, test condition A, t = 240 hours	Method 1042 of MIL-STD-750, test condition A; or t = 48 hours minimum at +175°C min
13	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1} and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1}, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100 percent probed in accordance with table I, subgroup 2 except test current shall not exceed 20 amperes.

4.3.2 Gate stress test. Apply V_{GS} = 30 V minimum for t = 250 µs minimum.

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4.3.3 Single pulse unclamped inductive switching.

a. Peak current, I_D .....Rated I_{D1}.

b. Peak gate voltage, V_{GS} .....10 V.

c. Gate to source resistor, R_{GS} .....$25 \leq R_{GS} \leq 200$.

d. Initial case temperature .....+25°C +10, -5°C.

e. Inductance .....100 µH minimum.

f. Number of pulses to be applied .....1 pulse minimum.

g. Supply voltage V_{DD} .....50 V.

* 4.3.4 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW}, (and V_H where appropriate). (See figure 5 herein.) See table II, group E, subgroup 4 herein.

* 4.3.5 PIND. Not applicable in screening when devices are processed using alternative method and flow requirements approved by the qualifying activity, that includes incorporating the use of certified clean processing and silicone die coat. Instead, the PIND test performance shall be performed in group B3 and group C3, on a lot sample basis. PIND failures detected in group B or C will represent lot jeopardy and shall be evaluated for root cause and lot integrity.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

	Subgroup	Method	Conditions
	B3	1051	Test condition G.
*	B3	2052	PIND, required if not performed in screening. (22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	B4	1042	Test condition D; the heating cycle shall be 1 minute minimum.
	B5	1042	Accelerated steady-state operation life; test condition A, V_{DS} = rated T_A = +175°C, t = 120 hours minimum. Read and record $V_{(BR)DSS}$ (pre and post at 1 mA = I_D. Read and record I_{DSS} (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10 percent and I_{DSS} shall not exceed 25 µA. Accelerated steady-state gate stress; condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.
	B5	2037	Bond strength; test condition D.

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4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	1051	Test condition G.
B3	1042	Test condition D. The heating cycle shall be 1 minute minimum.
B3	2037	Test condition A. All internal bond wires for each device shall be pulled separately.
B4	2075	See 3.4.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500.

	Subgroup	Method	Conditions
	C2	1056	Test condition A.
*	C2	2036	Test condition A; weight = 10 lbs. (4.54 Kg), t = 15 s.
*	C3	2052	PIND, required if not performed in screening. (JANS only, 22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	C5	3161	See 4.3.4,
	C6	1042	Test condition D. The heating cycle shall be 1 minute minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
* Thermal impedance 2/	3161	See 4.3.4	$Z_{\theta JC}$			°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = 1 mA dc, condition C	$V_{(BR)DSS}$			
2N6756				100		V dc
2N6758				200		V dc
2N6760				400		V dc
2N6762				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 V dc, V_{DS} = 0, bias condition C	I_{GSSF1}		± 100	nA dc
Gate current	3411	V_{GS} = -20 V dc, V_{DS} = 0, bias condition C	I_{GSSR1}		± 100	nA dc
Drain current	3413	V_{GS} = 0 V dc V_{DS} = 80 percent of rated V_{DS}; bias condition C	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1); condition A, I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			
2N6756					0.18	ohms
2N6758					0.4	ohms
2N6760					1.0	ohms
2N6762					1.5	ohms
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1); condition A, I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			
2N6756					.21	ohms
2N6758					.49	ohms
2N6760					1.22	ohms
2N6762					1.80	ohms

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward voltage (source drain diode)	4011	Condition A, pulsed (see 4.5.1), V_{GS} = 0 V, I_D = I_{D1}	V_{SD}			
2N6756					1.8	V dc
2N6758					1.6	V dc
2N6760					1.5	V dc
2N6762					1.4	V dc
Subgroup 3						
High temperature operation:		T_C = T_J = +125°C				
Gate current	3411	Bias condition C V_{GS} = ± 20 V dc V_{DS} = 0 V dc	I_{GSS2}		± 200	nA dc
Drain current	3413	Bias condition C V_{GS} = 0 V dc V_{DS} = 100 percent of rated V_{DS}	I_{DSS2}		1.0	mA dc
		V_{DS} = 80 percent of rated V_{DS}	I_{DSS3}		0.25	mA dc
Static drain to source on-state resistance	3421	Condition A, V_{GS} = 10 V dc pulsed, (see 4.5.1); I_D = rated I_{D2}	$r_{DS(on)3}$			
2N6756					0.34	ohms
2N6758					0.8	ohms
2N6760					2.2	ohms
2N6762					3.3	ohms
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)2}$	1.0		V dc
Low temperature operation:		T_C = T_J = -55°C				
Gate to source voltage threshold	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)3}$		5.0	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1}; V_{GS} = 10 V dc Gate drive impedance = 7.5Ω V_{DD} = 50 percent of $V_{BR(DSS)}$				
Turn-on delay time			$t_{d(on)}$			
2N6756, 2N6758					35	ns
2N6760, 2N6762					30	ns
Rise time			t_r			
2N6756, 2N6758					80	ns
2N6760, 2N6762					40	ns
Turn-off delay time			$t_{d(off)}$			
2N6756, 2N6758					60	ns
2N6760, 2N6762					80	ns
Fall time			t_f			
2N6756					45	ns
2N6758					40	ns
2N6760					35	ns
2N6762					30	ns
Subgroup 5						
Safe operating area test	3474	See figure 6; V_{DS} = 80 percent of rated $V_{BR(DSS)}$, V_{DS} = 200 V max; t_p = 10 ms				
Electrical measurements		See table I, subgroup 2				
Single pulse unclamped inductive switching	3470	See 4.3.3, c = 0, 116 devices				
Electrical measurements		See table I, subgroup 2				
Subgroup 6						
Not applicable						

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Gate charge	3471	Condition B				
On-state charge			$Q_{g(on)}$			
2N6756					35	nC
2N6758					39	nC
2N6760					39	nC
2N6762					40	nC
Gate to source charge			Q_{gs}			
2N6756					10	nC
2N6758					5.7	nC
2N6760, 2N6762					6.0	nC
Gate to drain charge			Q_{gd}			
2N6756					15	nC
2N6758					20	nC
2N6760					20	nC
2N6762					20	nC
Reverse recovery time	3473	Condition A, di/dt = 100 A/µs $V_{DD} \leq 30$ V, $I_D = I_{D1}$	t_{rr}			
2N6756					300	ns
2N6758					500	ns
2N6760					700	ns
2N6762					900	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 6.
Group E, subgroup 1.

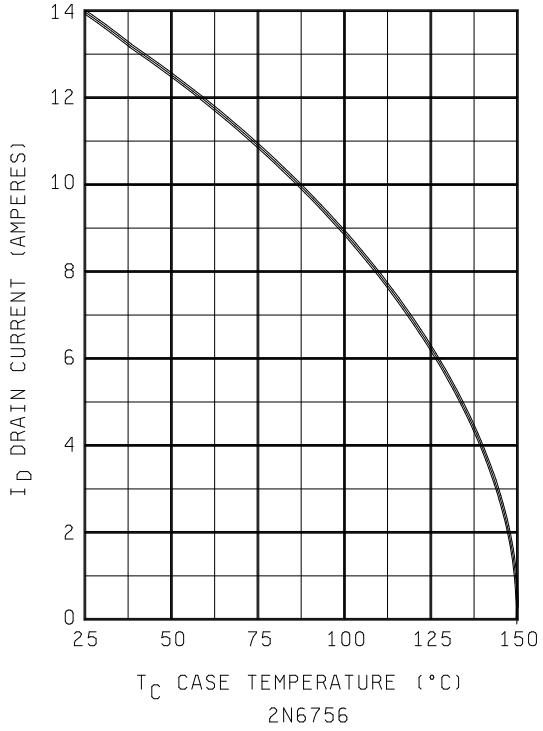
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TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

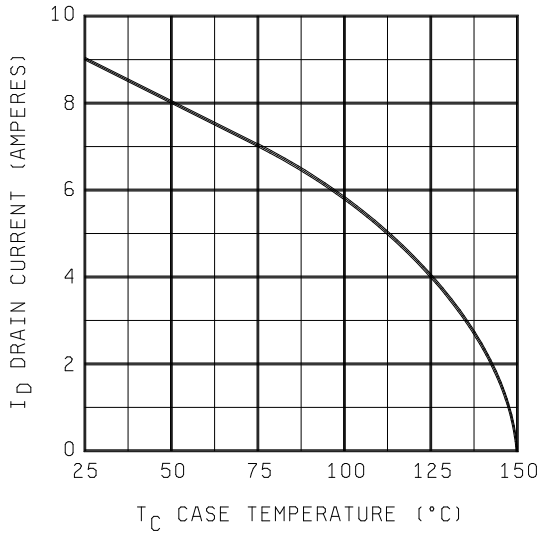
Inspection 1/	MIL-STD-750		Qualification and large lot quality conformance inspection 1/
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycle	1051	Condition G, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2	
Subgroup 2 2/			45 devices c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Steady-state gate bias	1042	Condition B, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			sample size N/A
Thermal impedance curves		See MIL-PRF-19500	
Subgroup 5			3 devices c = 0
Barometric pressure (reduced) 400 and 500 V only	1001	Test condition C $V_{ISO} = V_{DS}$, $I_{(ISO)}$ = .25 mA (max)	
Subgroup 10			
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer	22 devices c = 0

1/ JANHC and JANKC devices are qualified in accordance with appendix G of MIL-PRF-19500.
2/ A separate sample for each test may be pulled.

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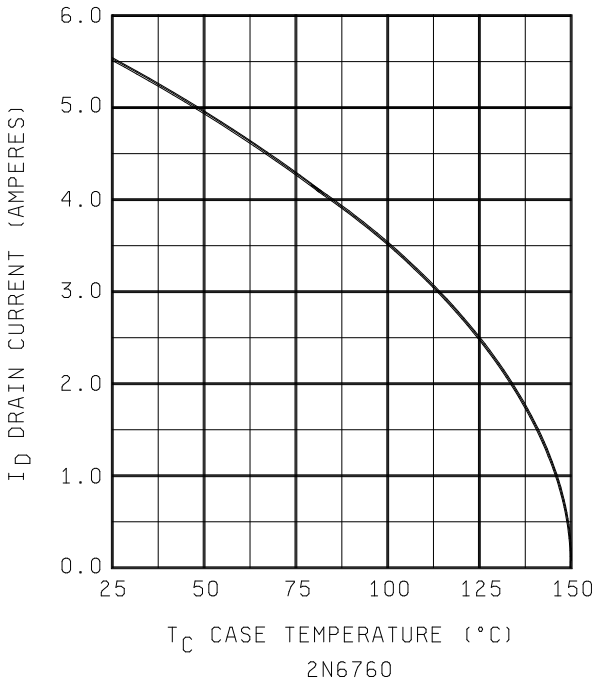
2N6756



2N6758

FIGURE 4. Maximum drain current versus case temperature.

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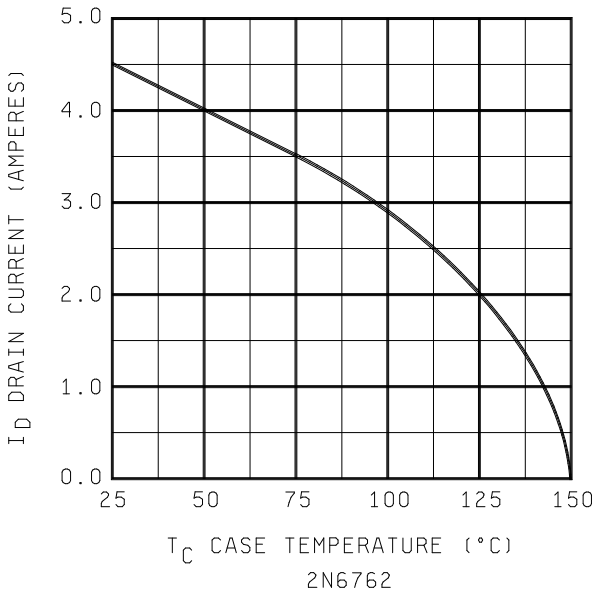


FIGURE 4. Maximum drain current versus case temperature - Continued.

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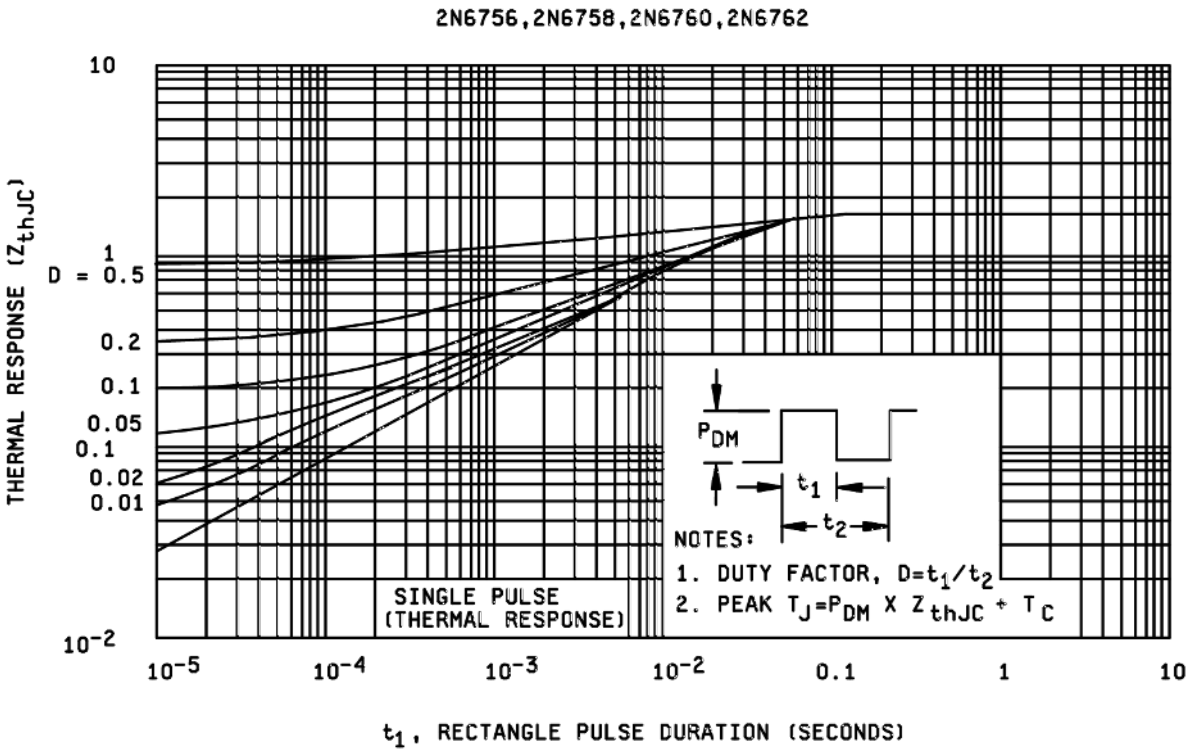


FIGURE 5. Thermal response curves.

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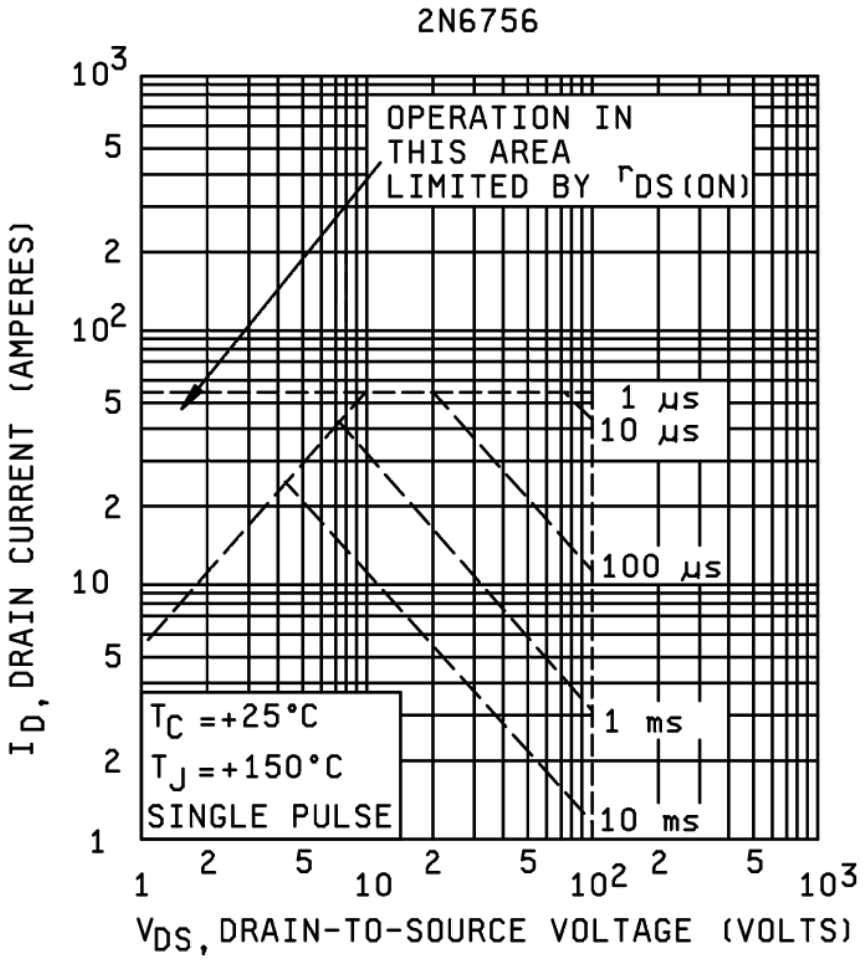


FIGURE 6. Safe operating area.

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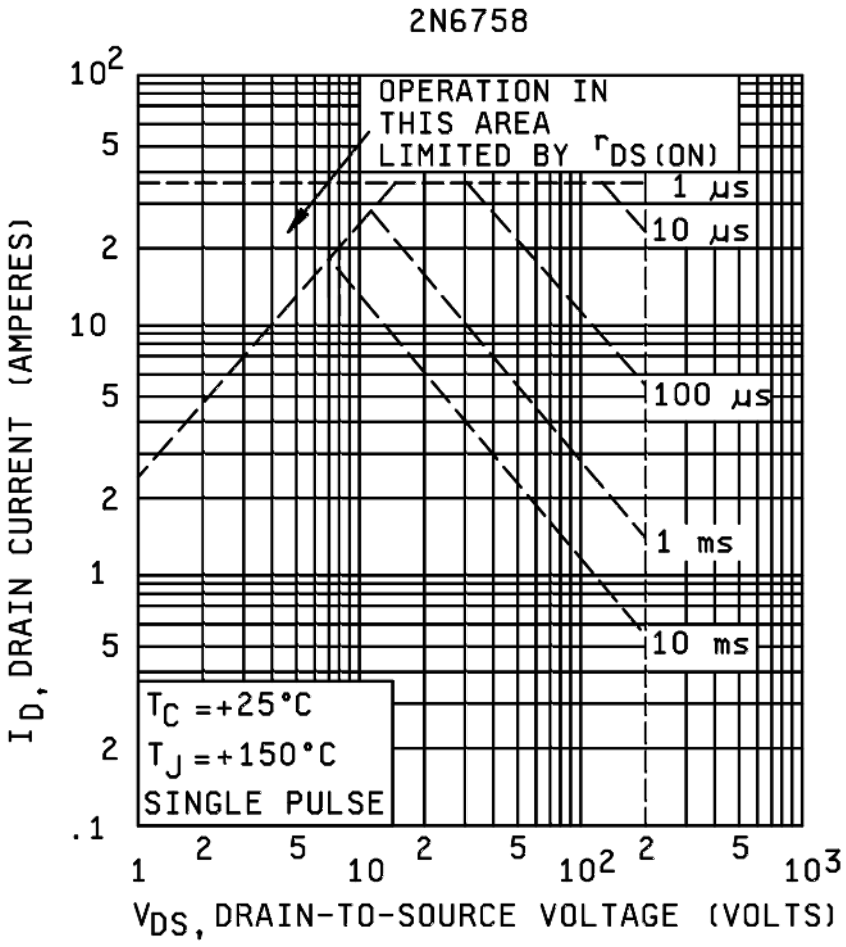


FIGURE 6. Safe operating area - Continued.

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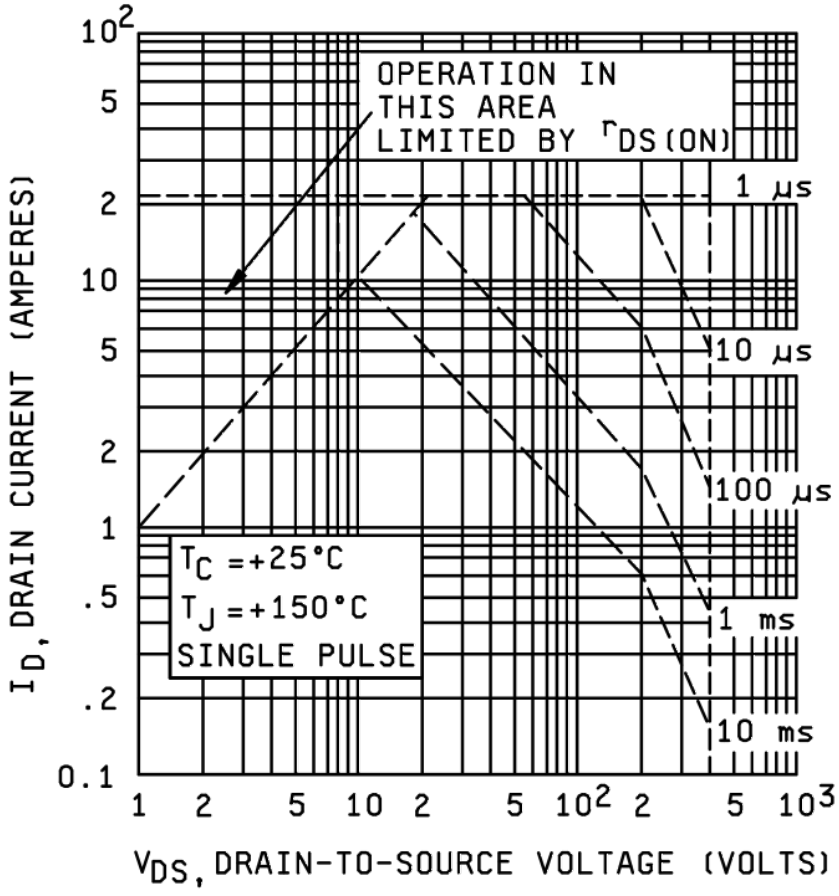


FIGURE 6. Safe operating area - Continued.

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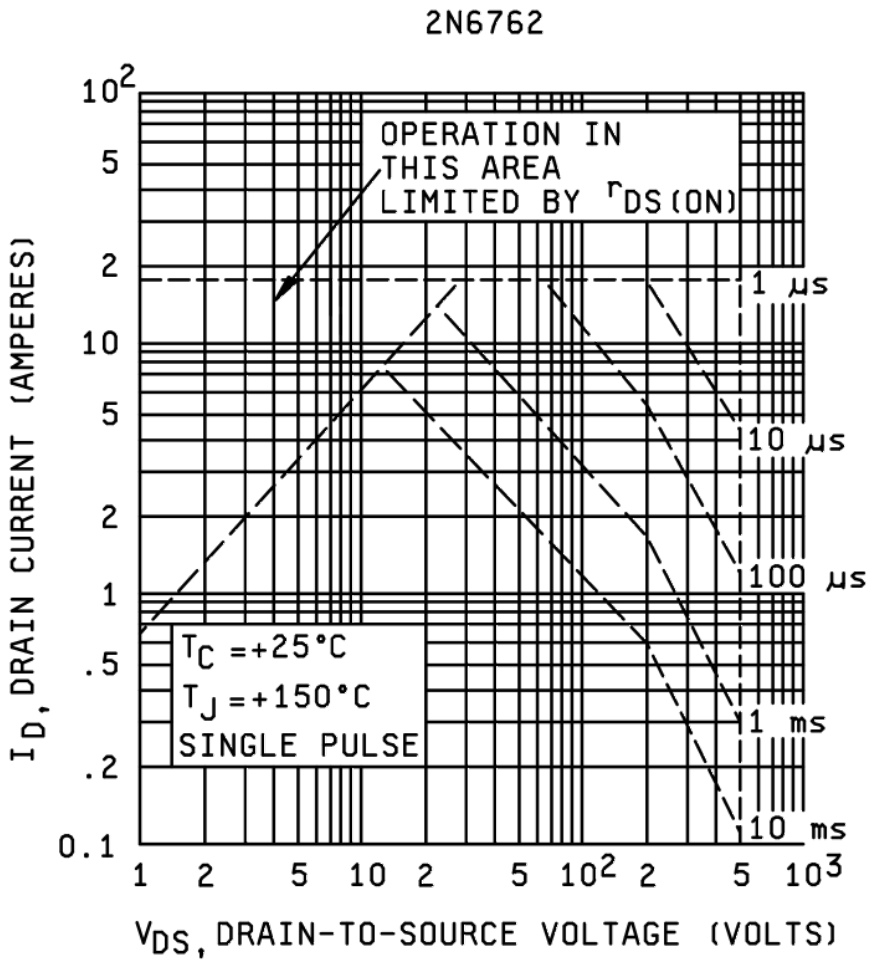


FIGURE 6. Safe operating area - Continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.6.

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 2 and 3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

* 6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

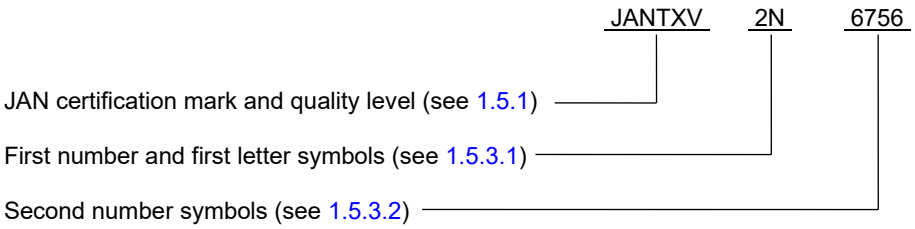
PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN
2N6756	69210	IRF130
2N6758	69210	IRF230
2N6760	69210	IRF330
2N6762	69210	IRF430

6.5 Replacement data. JANTX devices shall be a direct replacement for JAN devices (example: JANTX2N6756 for JAN2N6756).

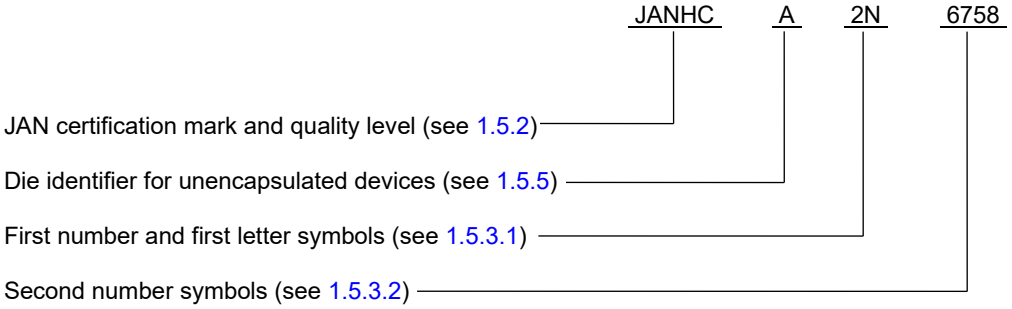
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6.6 PIN construction example.

6.6.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



6.6.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.7 List of PINs.

6.7.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N6756	JANTX2N6756	JANTXV2N6756	JANS2N6756
JAN2N6758	JANTX2N6758	JANTXV2N6758	JANS2N6758
JAN2N6760	JANTX2N6760	JANTXV2N6760	JANS2N6760
JAN2N6762	JANTX2N6762	JANTXV2N6762	JANS2N6762

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* 6.8 List of PINs for unencapsulated devices. The following is a list of possible PINs available on this specification sheet. The qualified die suppliers with the applicable letter version (example JANHCA2N6756) will be identified on the QML.

JANC ordering information		
PIN	Manufacturers	
	69210	43611
2N6756	JANHCA6756 JANKCA6756	JANHCC2N6756 JANKCC2N6756
2N6758	JANHCA6758 JANKCA6758	JANHCC2N6758 JANKCC2N6758
2N6760	JANHCA6760 JANKCA6760	
2N6762	JANHCA6762 JANKCA6762	

6.9 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

* 6.10 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Navy - AS
* Air Force - 19, 70, 170

Preparing activity:
DLA - CC

(Project 5961-2021-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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